

— PROPOSAL • SMART GEYSER AND SOLAR PV

Solar Hot Water for 136 Homes at Scottsdene

Supply, installation, testing, commissioning and certification of a site-level grid-tied solar PV system, and installation and commissioning of a Plentify-supplied HotBot on every geyser, sized to offset at least 75% of annual hot water heating energy, handed over before 1 November 2026.

PREPARED FOR PLENTIFY, FOR INTERNATIONAL HOUSING SOLUTIONS

OUR REFERENCE

SL-2609-PLY

DATE

2 October 2026

COMPLETION

Before 1 November
2026

VALID

30 days

— COVERING LETTER

What We Are Proposing

Attention: Plentify

Re: Smart Geyser and Solar PV Solution for IHS, Scottsdene Residential Development, Cape Town.

Thank you for the request for quotation. We have worked from the RFQ, the 2019 council plans, the signed deviation council pack of 20 May 2026, the site development plan (SD-P9B-E3830 Rev E), the schedule of finishes and the site photographs of June and August 2026. Where the deviation pack changed the design, we have followed it.

We propose one solar system for the whole development: 404 LONGi 615 W bifacial modules on the existing tiled roofs feeding two three-phase grid-tied inverters, each with a Plentify SolarBot, at zero export and with no batteries. All 136 geysers move to a central, landlord-metered geyser supply built by the site electrical contractor, and a Plentify HotBot on every geyser schedules heating against the site's solar output. The geysers are the store.

The roof sets what is possible here. The deviation pack changed the roofs from mono-pitch zinc to dual-pitch concrete tile at 20°, 8.69 m wide on blocks 1 to 3 and 6.95 m on block 4, with the stair cores under flat concrete slabs. Our array layout is traced from the roof plans in that pack and fits 404 modules on the tiles. With one pool of solar and 136 geysers to heat in turn, our model loses little midday output to full tanks and puts the development at about 82% annual offset against the 75% target. The final rational calculation will be made on surveyed roof faces.

The drawings already commit the project to 50% of hot water from solar-assisted heating under SANS 10400-XA. This system is designed to meet that, and the RFQ's higher 75% target.

Moving the geysers off the tenants' prepaid meters means the savings reach tenants through billing, using HotBot's per-geyser metering, or through hot water included in the rent. That choice sits with IHS, and we set out what each needs.

Your contact for everything on this proposal is **Stephan Marais** (stephan@sunlogic.co.za). He handles the PV design, the site, the PV certification and the handover, and you will not be passed between people.

— THE SHORT VERSION

404 modules on two inverters, 136 HotBots, zero export and no batteries clear the 75% target.

The geysers move to a central landlord-metered supply, so IHS decides how tenants receive the savings.

How the System **Works**



One Solar Pool

404 LONGi 615 W bifacial modules (about 248 kWp) on the tiled roofs, collected by roof face into a DC combiner per block and fed to two Deye SUN-110K-G03 110 kW three-phase grid-tied inverters.



Smart Control Throughout

A Plentify SolarBot and a Deye meter on each inverter hold the site at zero export. A Plentify HotBot on every geyser heats the 136 tanks in turn to use the solar as it arrives. Plentify supplies both.



Measured, Not Estimated

Both inverters report to our VoltIQ monitoring platform, alongside Plentify's platform for SolarBots and HotBots, so the offset is shown rather than claimed.

Two Inverters, **One Geyser Supply**

WHY TWO INVERTERS

248 kWp of modules on two Deye 110 kW string inverters gives a DC-to-AC ratio of about 1.13. With six MPPT inputs per inverter, every roof face gets its own input, half the solar output continues if one fails, and both are expected to sit within the SolarBot-supported range, subject to Plentify's confirmation. A single inverter of that size is possible only if Plentify confirms SolarBot supports it.

THE CENTRAL GEYSER SUPPLY

A main geyser board with a landlord bulk meter at the inverter room, sub-mains to a geyser board in each block, all by the site electrical contractor. The electrical contractor also reroutes each geyser circuit from the unit board in the service duct. Done now, while the blocks are open, this is simpler and cheaper than after occupation.

TENANT SAVINGS

Either tenants are billed only for the grid energy their own geyser draws, from HotBot's per-geyser metering where it is accepted for billing, or hot water is included in the rent or levy. IHS chooses; both are compatible with the system as designed.

GRID OUTAGES

Grid-tied inverters shut down when the grid is off, as the grid code requires. During load-shedding there is no solar heating; HotBot reheats once the grid returns. The final calculation deducts expected outage hours.

What the Roof **Allows**



Four blocks, two roof widths

Four blocks of four storeys hold 136 units, 34 per floor across the site: 135 two-bed and one one-bed. Every roof is dual-pitch concrete tile at 20° on SA pine trusses. Blocks 1 to 3 are 8.69 m wide; the three wings of block 4 are 6.95 m wide. The stair cores sit under flat concrete slabs with 170 mm parapets, set into the courtyard side of each roof.



404 modules, placed on the drawn roofs

Each module is placed within the roof outlines taken from the linework of the deviation pack, clear of the ridges and stair-core slabs, with 0.35 m clear at the eaves, 0.3 m at the ridge and 0.4 m at gables and slabs. The 8.69 m roofs take three landscape rows per face; block 4's 6.95 m roofs take two. The flat slabs stay in reserve.



About 82% annual offset

True north sits about 35° off the site plan. Blocks 1 to 3 face north-east and south-west, with north-north-west and south-south-east faces on their short wings; block 4's diagonal wing faces east and west. In our model, October to March is fully covered and June falls to about 48%. If vents and wind-zone edges cost 10% of the modules, the offset is still about 78%.



Fitted to a finished roof

The August 2026 photographs show the roofs tiled and scaffolding still up on the long blocks. We fix tile hooks to the trusses beneath the tiles, share the scaffolding while it stands, and run cables down the service ducts before their steel doors are fitted. Every unit is separated by a firewall sealed to the underside of the tiles; cables cross them only through fire-rated sleeves.

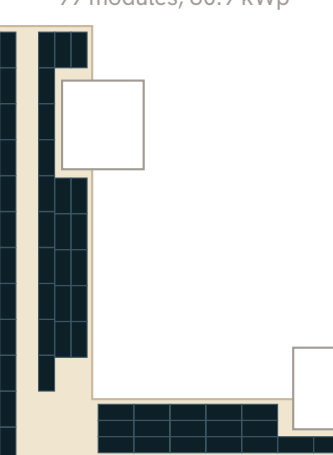


The geyser supply is the decision

Moving 136 geysers to a landlord bulk supply needs IHS's agreement, a supply sized with HotBot load control, and the City's approval of the supply, applied for by the site electrical contractor. The bulk supply, and any upgrade to it, is by the site electrical contractor and the City, outside this price.

Block 1

99 modules, 60.9 kWp



A 3D architectural rendering of a building facade, identified as 'Block 1'. The facade is composed of dark blue solar panels and light beige concrete. The building has a stepped profile with four distinct levels. Two white rectangular boxes highlight specific areas: one on the upper left and one on the lower right. The text 'Block 1' is displayed in bold black font, and '99 modules, 60.9 kWp' is displayed in a smaller black font below it.

North-east	110
South-west	141
North-north-west	51
West	32
East	32
South-south-east	38

-  PV module, landscape, 2 382 × 1 134 mm
-  20° concrete tile roof
-  Flat concrete slab (stair core)

The Priced Schedule

Two site inverters, a central landlord-metered geyser supply and a HotBot on every geyser.

1 Preliminary & General

Establishment, supervision, transport, scaffolding, lifting and safety for the duration of the works.

DESCRIPTION	QTY	RATE	TOTAL
Site establishment	1 ITEM	R 3 000	R 3 000
Supervision	1 ITEM	R 15 000	R 15 000
Transport	1 ITEM	R 10 500	R 10 500
Access tower, about 13 m, rental including erect and dismantle, 5 weeks, per block	4 NO	R 18 750	R 75 000
Eaves edge protection, rental with erection, about 120 m moved between blocks, 5 weeks	1 ITEM	R 60 000	R 60 000
Telehandler rental with operator, 14 to 17 m reach, including delivery	15 DAYS	R 4 000	R 60 000
Safety harness and temporary lifeline rental, 8 harnesses and 4 lifelines	5 WEEKS	R 8 000	R 40 000
Project management, health and safety file, handover documentation	1 ITEM	R 20 000	R 20 000
SECTION 1 SUBTOTAL			R 283 500

2 Engineering & Design

Professional sign-off of the roof structure and the PV electrical design, its interface with the central geyser supply, and the final offset calculation.

DESCRIPTION	QTY	RATE	TOTAL
Structural engineer: roof inspection, load check, mounting sign-off	1 ITEM	R 30 000	R 30 000
Electrical engineer: PV design and its interface with the central geyser supply, offset calculation, single-line diagrams, City SSEG application	1 ITEM	R 30 000	R 30 000
SECTION 2 SUBTOTAL			R 60 000

3 PV Modules & Mounting

LONGi 615 W bifacial monocrystalline modules, 2 382 × 1 134 mm, on tile hooks and rails fixed to the trusses of the 20° concrete tile roofs, engineered to the verified structure.

DESCRIPTION	QTY	RATE	TOTAL
PV module, LONGi 615 W bifacial	404 NO	R 1 690	R 682 760
Tile-roof mounting, hooks and rails, per module	404 NO	R 400	R 161 600
PV module installation labour, per module	404 NO	R 900	R 363 600
SECTION 3 SUBTOTAL			R 1 207 960

4 DC Collection

Strings grouped by roof face, a DC combiner per block with fuses, surge protection and isolation, and DC trunk cabling to the inverter room. Electrical items are subject to the site electrical contractor's final design; changes they make may change this price.

DESCRIPTION	QTY	RATE	TOTAL
PV string cabling and connectors, roof to combiner	1 ITEM	R 83 440	R 83 440
DC combiner box per block, fused, with surge protection and isolator	4 NO	R 11 700	R 46 800
DC trunk cable, combiner to inverter room	280 METRE	R 545	R 152 600
Earthing and lightning protection bonding to arrays, subject to the electrical contractor's design	1 ITEM	R 42 000	R 42 000
SECTION 4 SUBTOTAL			R 324 840

5 Inverters & Control

Two Deye SUN-110K-G03 110 kW three-phase grid-tied string inverters, each with a Plentify SolarBot (supplied by Plentify, fitted with the inverter), a Deye 250 A meter with CT for zero-export control, and AC protection in a ventilated inverter room or enclosure. Electrical items are subject to the site electrical contractor's final design; changes they make may change this price.

DESCRIPTION	QTY	RATE	TOTAL
Inverter, Deye SUN-110K-G03, 110 kW three-phase grid-tied	2 NO	R 70 188	R 140 376
Deye 250 A smart meter with CT, zero-export control, per inverter	2 NO	R 3 348	R 6 696
AC protection and inverter output switchgear, subject to the electrical contractor's design	1 ITEM	R 99 600	R 99 600
Inverter room or enclosure, ventilation and plinth	1 ITEM	R 69 000	R 69 000
SECTION 5 SUBTOTAL			R 315 672

6 Smart Geyser Control

Plentify HotBot on every geyser, supplied by Plentify, installed and commissioned by us. HotBot schedules all 136 geysers against the site's solar output.

DESCRIPTION	QTY	RATE	TOTAL
Plentify HotBot, install and commission	136 NO	R 400	R 54 400
SECTION 6 SUBTOTAL			R 54 400

7 Monitoring & Servicing

VoltIQ monitoring of both inverters and the site offset, with onboarding and the first year of platform access; and the first year of OEM and annual servicing, which helps keep manufacturer warranties valid.

DESCRIPTION	QTY	RATE	TOTAL
VoltIQ onboarding and first-year access, site	1 ITEM	R 15 000	R 15 000
OEM and annual servicing, first year: inspection, module cleaning, inverter OEM service, SolarBot and HotBot checks with Plentify, tests and report	1 YEAR	R 60 000	R 60 000
SECTION 7 SUBTOTAL			R 75 000

8 Testing, CoC & Registration

Testing and commissioning, a Certificate of Compliance for the PV installation, and registration of the system with the City of Cape Town. Electrical items are subject to the site electrical contractor's final design; changes they make may change this price.

DESCRIPTION	QTY	RATE	TOTAL
Full electrical test and CoC, PV installation	1 ITEM	R 1 500	R 1 500
City of Cape Town SSEG application and registration	1 ITEM	R 7 500	R 7 500
SECTION 8 SUBTOTAL			R 9 000

Rates are complete, inclusive of equipment and labour. The DC trunk cable length is estimated from the drawings, re-measured on site and adjusted at the quoted rate per metre. Sunlogic is the solar supplier and installer only. The site's electrical contractor remains responsible for the building's electrical installation; any change they require to the design, cable routes, boards or protection may change this price, and we will price it and agree it with you before acting on it.

Provisional Sums

P1 • STAIR-CORE SLAB FRAMES

Low-tilt frames on the flat stair-core slabs, only if the survey finds the tile faces take fewer than 404 modules.
Measured and agreed before ordering.

— SUMMARY

What It Comes To

SECTION A, FIRM PRICE EXCLUDING VAT

R 2 330 372.00

SECTION B, PROVISIONAL SUMS

Measured and agreed

VAT @ 15%

R 349 555.80

TOTAL INCLUDING VAT

R 2 679 927.80

About R 17 135 per unit and R 9 379 per kWp, excluding VAT. Electrical items depend on the site electrical contractor's final design and may change with it.

— PROGRAMME

From Award to Handover

- **2 to 5 October.** Site and roof survey, run during the tender at our risk.
- **9 October.** Design freeze: structural sign-off, array layout, geyser supply interface with the electrical contractor and offset calculation approved by Plentify; City application submitted.
- **15 to 29 October.** Rollout, block by block: roof arrays, DC combiners and HotBot installation in four parallel crews, Saturdays 17 and 24 October worked.
- **Saturday 31 October.** Handover file: as-built drawings, test sheets, warranties and a tenant information leaflet.
- **6 October.** Award and deposit; modules and inverters ordered the same day, subject to supplier stock at award.
- **12 to 14 October.** Two pilot units installed and approved by Plentify and IHS.
- **30 October.** CoC issued for the PV system, both inverters and all HotBots online, City registration in progress.
- **One day of float.** Each day award slips beyond 6 October needs a fifth crew on top of the four, or a later date agreed with IHS.

What This Price Assumes

If any of these turns out not to hold, we will tell you before we act on it, not afterwards.

- ✓ Sunlogic is the solar supplier and installer only. The site's electrical contractor remains responsible for the building's electrical installation; any change they require to the design, cable routes, boards or protection may change this price, and we will price it and agree it with you before acting on it.
- ✓ Award and deposit by 6 October 2026, with roof access for the survey from 2 October.
- ✓ The tiled roofs take 404 modules as drawn in the array layout.
- ✓ Units are unoccupied during installation, with free access to roofs and geyser ducts.
- ✓ Cables cross unit firewalls only through fire-rated sleeves, to the architect's approval.
- ✓ The existing site connection, with HotBot load control, can carry the geyser supply; if not, the upgrade is by the site electrical contractor and the City, outside this price.
- ✓ Space is provided for an inverter room or enclosure near the LV kiosk.
- ✓ The gatehouse bachelor flat added in the deviation pack is not one of the 136 units and is outside this scope unless instructed.
- ✓ An internet connection at the inverter room (a LAN point or a data SIM) for the two inverter data loggers that report to VoltIQ is provided by IHS.
- ✓ The site electrical contractor supplies and installs the main geyser board with bulk meter, the sub-mains and the four block geyser boards and reroutes and certifies all 136 geyser circuits, in time for HotBot commissioning.
- ✓ The geysers are new Kwikot Superline 150 L units, compliant, with valves and ducts in working order.
- ✓ The trusses carry the added load of about 15 kg/m² without strengthening, subject to the engineer.
- ✓ Our scaffolding and lifting are a fixed sum. Where the main contractor's scaffolding is still up we use it as well, by agreement and at no charge either way. Conduits and cables go into the service ducts before their steel doors are fitted.
- ✓ IHS accepts the geysers on a landlord-metered supply and decides how tenants are credited.
- ✓ SolarBot supports the selected three-phase grid-tied inverter, as Plentify is to confirm; if not, an alternative inverter is priced and agreed before ordering. Plentify supplies the HotBots and SolarBots, and their connection to Plentify's platform; no site network is included.
- ✓ IHS signs the City of Cape Town application as owner.
- ✓ The DC trunk cable length is estimated from the drawings; it is re-measured on site and adjusted at the quoted rate per metre.

What Is Excluded

- **The building's electrical installation.** Supply boards, unit wiring and any work the site electrical contractor carries out or changes. We price only the solar supply and installation, and the interfaces listed in the schedule.
- **Common areas and batteries.** Per the RFQ, only residential geysers are served, and there is no storage beyond the geysers.
- **Geysers, plumbing and roofing.** Replacement geysers, plumbing repairs and roof repairs not caused by our work.
- **Ongoing costs.** Maintenance, network data and VoltIQ fees after the first year, unless the annual service is renewed.
- **Central geyser supply and geyser circuits.** The main geyser board with landlord bulk meter, the sub-mains, the block geyser boards, the rerouting of all 136 geyser circuits and their CoCs are by the site electrical contractor.
- **HotBots, SolarBots and connectivity.** Supplied by Plentify, with their connection to Plentify's platform. We install and commission them; no site Wi-Fi network is included.
- **Roof strengthening.** Strengthening or alternative mounting, if the structural engineer requires it, is priced separately once measured.
- **Tenant billing.** Setting up or running tenant billing for geyser energy; we provide the per-geyser data through HotBot.

Commercial Terms

VALIDITY

30 days from the date of this proposal.

PAYMENT

50% on award, to secure modules and inverters for the October programme. 40% on delivery of materials to site. 10% on handover of the PV CoC and the handover file.

WARRANTIES

Modules, inverters, SolarBots and HotBots carry their manufacturers' warranties. Workmanship and roof waterproofing: 12 months from handover.

CERTIFICATION

A Certificate of Compliance for the PV installation, issued by a registered electrician, and registration of the system with the City of Cape Town. Geyser circuit CoCs are issued by the site electrical contractor.

Annual Service and Monitoring

WHY IT MATTERS

Manufacturer warranties on modules, inverters, SolarBots and HotBots usually need documented annual servicing. It helps keep them valid and supports the offset after handover. The first year is included in the schedule; renewal is offered yearly after that.

MONITORING

VoltIQ monitoring of both inverters and the site offset, fault alerts and a monthly offset report to IHS.

MAINTENANCE

Annual roof and module inspection, DC connection and isolator checks, insulation resistance tests, module cleaning, and re-sealing of roof penetrations where needed.

OEM SERVICING AND CALL-OUTS

Inverter firmware and function tests, SolarBot and HotBot servicing with Plentify, warranty claims with each manufacturer, and call-outs, with response times agreed at renewal. An annual performance report is included.

Your Contact Person



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Solar PV and smart energy systems for homes,
businesses and developments in Cape Town.

— NEXT STEP

Confirm the Geyser Supply and We Start the Survey

The roof survey can begin on 2 October, ahead of award, so the programme holds.



Solar PV and smart energy systems for
homes, businesses and developments
in Cape Town.

THIS DOCUMENT

SL-2609-PLY

Scottsdene Residential
Development, Cape Town

Issued 2 October 2026 ·
valid 30 days

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